

Study on the reignition phase and magnetic field simulation of drawn vacuum arc in 126 kV axial magnetic field contacts

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ABSTRACT

With the increase in breaking voltage level, studying the evolution characteristics of reignition of high-current vacuum arc plays an important role to improve the breaking capacity of vacuum circuit breakers. In order to analyze the characteristics of the arc in the reignition phase of 126 kV axial magnetic field (AMF) contacts, drawn vacuum arc experiments were conducted in a detachable vacuum chamber and magnetic field simulations were performed under corresponding conditions. By analyzing the arc images and arc-voltage characteristic of high-current interruption experiments with a pair of 140-mm-diameter cup-type six-slot AMF contact, the arcing reignition process in the second half-wave can be divided into four stages, i.e., reignition stage, disordered movement stage, dispersed arc stage, transition stage, and diffused arc stage. Combined with the magnetic field simulation analysis, at the beginning of the second half-wave just after current zero, the axial magnetic field is weak, resulting in poor control over the arc, while the transverse magnetic field is strong, promoting rotational motion of the arc along the contact surface. In this process, the arc voltage presents significant noise. As the current increases and the axial magnetic field strengthens, the control of the arc gradually improves. The arc transitions from dispersed mode to diffused mode after the current peak. At the same time, cathode spots continuously spread on the contact surface, the arc voltage gradually becomes stable, and the arc extinguishes at the second current zero.

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I. INTRODUCTION

In recent years, significant achievements have been made in the development of vacuum circuit breakers (VCBs), which is primarily attributed to their environmental friendliness, compact size, and reliable performance in current interruption. Initially designed for medium-voltage applications, VCBs have gradually expanded their application scope, including widespread use in power systems up to 126 kV and even higher voltage levels.

Early research on VCBs primarily focused on the interruption at the medium-voltage level, with a multitude of experiments conducted to understand the interruption process, including the ignition, extinction, and reignition of the arc. These studies have provided valuable insights into the behavior of vacuum arcs under different operating conditions. Feng *et al.* utilized high-speed photography to observe the arc ignition process in the vacuum contact gap, revealing the micro-discharge phenomena and the formation mechanism of arc initiation. They further investigated the relationship between the number of

initial arc columns and parameters such as the initial arc voltage and initial arc current.¹ Chaly *et al.* conducted experimental research on the arc extinction characteristics under different magnetic field conditions. They found that an axial magnetic field (AMF) could effectively enhance the arc extinction speed and reduce the possibility of arc reignition.² Jiang *et al.* conducted in-depth research on the mechanism of arc reignition and discovered that the re-evaporation of metal vapor from the contact surface is the main reason leading to arc reignition and proposed a method to suppress reignition by optimizing the contact material and shape.³ With the increasing demand for higher breaking capacity, the focus of research has shifted to interruption experiments with high current and large contact gap, particularly at the 126 kV voltage level.

At this higher voltage level, interrupting fault currents faces greater challenges due to the increase in arc energy and the complexity of arc behavior. The development of axial magnetic field (AMF) contacts is a significant advancement in the technology of vacuum circuit

breakers. AMF contacts control the vacuum arc by applying a magnetic field along the axis of the contacts, which helps to stabilize the arc and facilitate its extinction. Li *et al.* have investigated the impact of AMF on arc stability through experiments and simulations. They found that the AMF could effectively compress the arc, reduce its spread, thereby improving arc stability and decreasing arc erosion on the contacts.⁴ Liu *et al.* observed that the AMF could accelerate arc cooling and reduce thermionic emission, thereby promoting rapid arc extinction.⁵ Jiang and Li found that the geometric shape and material of the contacts significantly affected arc reignition and reduced the possibility of arc reignition through optimized design.⁶ Ju *et al.* investigated the free-burning characteristics of vacuum arcs under different gap distance-to-diameter ratio, providing theoretical support for further studies on arc control methods.⁷ Song *et al.* investigated the initial expansion process of drawn vacuum arcs and the influence of axial magnetic fields to deepen the understanding of the evolution mechanism of high-current vacuum arcs and provide a basis for vacuum interrupter design.⁸ Wang studied the characteristics of high-current vacuum arcs under AMF through numerical simulation, providing important theoretical foundations for the design and optimization of vacuum circuit breakers.⁹ Yang *et al.* explored the magnetic field characteristics between large-diameter cup-type AMF contacts under large-gap conditions, providing guidance and reference for the practical application of such contacts in high-voltage vacuum interrupters through numerical simulations.¹⁰

In conducting experiments related to AMF contacts, understanding the relationship between arc voltage and arc morphology is crucial for controlling the arc and improving the performance of VCB. Boxman and Sanders systematically analyzed the physical characteristics of vacuum arcs and pointed out that arc voltage is closely related to current density, electrode material, and gap distance. Under low-current conditions, the arc voltage is relatively low and stable, whereas under high-current conditions, the arc voltage significantly increases accompanied by severe fluctuation.¹¹ They also discovered that the diffuse arc corresponds to a lower and more stable arc voltage, whereas the constricted arc exhibits a higher voltage with greater fluctuations. The increase in arc voltage may be due to the arc constriction, leading to an increase in current density and energy concentration.¹² Tiwari studied the relationship between the temperature distribution of the arc plasma and the arc voltage, and they found that the increase in arc voltage is closely related to the localized concentration of plasma temperature.¹³

However, current research on the vacuum arc characteristics of AMF contacts at high-voltage levels (126 kV) primarily focuses on the first half-wave. There is limited study on the reignition phenomenon and mechanisms during the second half-wave, especially regarding the reignition characteristics of the drawn vacuum arc. The evolution of the arc in the second half-wave between large-diameter AMF contacts and large-gap distances remains unclear. To further understand the evolution characteristics of the vacuum arc after reignition in AMF contacts used in high-voltage level, this paper conducts drawn vacuum arc experiments on large-diameter AMF contacts and corresponding simulations of the magnetic field under the same operating conditions. This research will contribute to a deeper understanding of the reignition mechanisms in vacuum circuit breakers.

II. EXPERIMENTAL SETUP

The circuit topology, experimental setup, and measurement device are all shown in Fig. 1. The experimental circuit mainly consists

of an inductance–capacitance (LC) oscillating circuit with a frequency of 50 Hz and a detachable vacuum chamber. The LC circuit mainly includes a direct current (DC) power supply, a capacitor bank, an inductor, and a main circuit breaker. The detachable vacuum chamber encompasses the vacuum system and the operating mechanism. The measurement system is primarily composed of two high-speed cameras, a high-voltage probe, a Rogowski coil, an oscilloscope, and a computer.

During the experiments, the pressure in the detachable vacuum chamber is maintained between 10^{-4} and 10^{-3} Pa. The operating mechanism is a spring-driven type, which releases energy through the opening spring to pull the contacts apart, and the opening speed of the mechanism is adjusted by changing different springs. During the experiments, the opening speed of the operating mechanism is approximately 1.9 m/s during the first half-wave and about 2.0 m/s during the second half-wave. The evolution of the gap distance in the experiments is presented in Fig. 2.

In the experiments, two high-speed cameras were used to simultaneously record the ignition position and the complete evolution of the vacuum arc. As shown in Fig. 1, the two cameras observed the vacuum arc and cathode spots from two perpendicular directions. Camera A, a color camera, recorded the anode, arc column, and conductive rod region, and Camera B, a black-and-white camera, focused on the cathode region; their frame rates were 5000 and 100 000 fps, corresponding to 0.2 ms and $10\ \mu\text{s}$ per frame, respectively.

The moment when the current first flowed through the detachable vacuum chamber was defined as the global time reference, $T = 0$ ms. The oscilloscope was automatically triggered by the voltage change and recorded the corresponding current and arc-voltage waveforms, while both probes transmitted data to the oscilloscope for visualization and storage. The final synchronization between the cameras

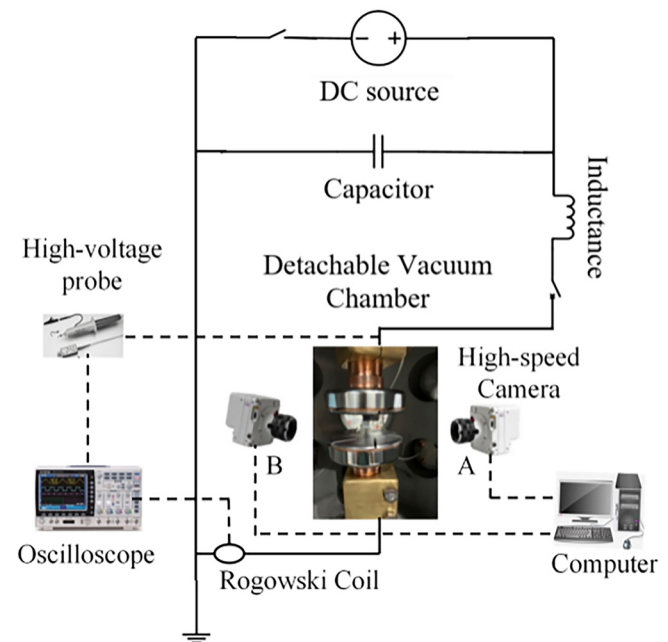


FIG. 1. Experimental setup.

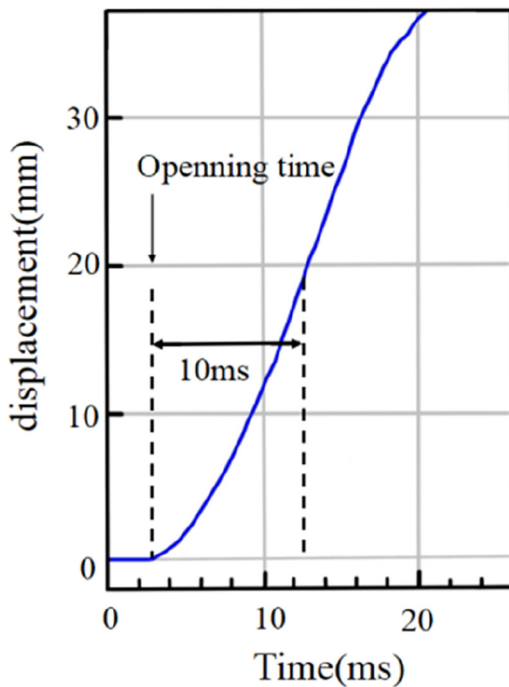


FIG. 2. The evolution of the gap distance.

and the oscilloscope was performed during post-processing. Specifically, for the image sequences recorded by the two high-speed cameras, the first frame in which visible light emission from the arc appeared was taken as a common reference frame for both views. This ignition frame corresponds to the instant of the abrupt voltage rise captured by the oscilloscope. Since Cameras A and B operated at fixed frame rates, the arcing time associated with each frame could be obtained by frame counting, allowing the arc images from both views to be accurately aligned with the corresponding voltage and current waveforms.

In the research related to 126 kV vacuum circuit breakers,^{14–18} cup-type contacts are typical structures of axial magnetic field contacts, which can effectively control and constrain the arc. The structures of the contacts used in the experiments are shown in Fig. 3. It is a six-slot cup-type contact with a diameter of 140 mm and a height of 40 mm.

The contact plates were made of CuCr50. Before collecting the interruption data summarized in Table I, each new pair of contacts was subjected to a series of conditioning operations. The conditioning current was set between 10 and 20 kA, which is lower than the typical current level at which reignition appears in our tests. The number of conditioning shots was not fixed but adjusted according to the observed arc behavior. The contacts were considered sufficiently conditioned when the following criteria were satisfied: (i) arc morphology stability—for a given current level, the overall appearance of the arc column and the distribution of cathode spots observed by the high-speed cameras became repeatable from shot to shot without sudden pattern changes, which are commonly seen on unconditioned contacts and (ii) voltage waveform stability—within the first half-wave, the arc-voltage waveform remained consistent over several consecutive

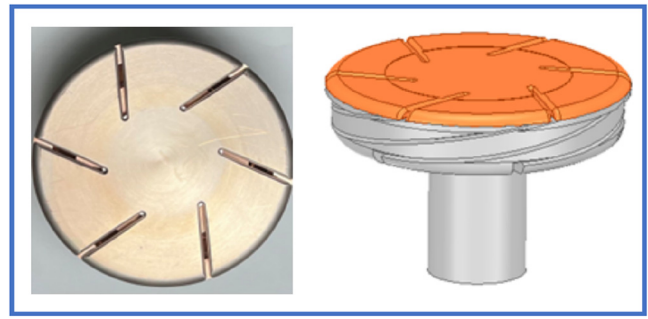


FIG. 3. Structures of six-slot cup-type contact.

operations without abnormal fluctuations or spurious extrema. Once these conditions were met, formal interruption experiments with peak currents from 10 to 30 kA were carried out, and for each current level, four repeated tests were performed.

In order to further understand the influence of the magnetic field on arc evolution during the second half-wave, the magnetic field distribution on the cathode surface of six-slot cup-type contacts at different contact gaps was simulated using ANSYS Maxwell software. In the simulation, the peak value of injected current I_{peak} was set to 30 kA with a frequency of 50 Hz, and the electromagnetic model was based on an eddy-current field. The arc morphology with side-jet characteristics during the second half-wave was also considered. Since the vacuum arc and contact gap vary over time, the magnetic field distribution was simulated by adjusting the contact gap and phase angle to represent different time points and the time step of the simulation was 0.1 ms. Some simulation parameters were derived from drawn vacuum arc experiments, such as a contact gap of 20 mm at current zero, and an average contact separation speed of 2 m/s was adopted to determine the change of gap during the simulation. The electrical conductivity of the vacuum arc was set to an effective uniform value of 2000 S/m, which is sufficient for resolving the concerned macroscopic B_z and B_t distributions mainly governed by the AMF contact geometry and current path, rather than by fine-scale conductivity variations within the arcing region.

III. EXPERIMENTAL RESULTS AND DISCUSSION

After sufficient conditioning of the contact surface, it was found that when using currents below 20 kA for breaking experiments, the AMF contacts could effectively control the arc, and no reignition occurred. Reignition only appeared in drawn arc experiments with peak current exceeding 20 kA. Therefore, only the experimental groups with current levels above 20 kA were included in the statistical analysis. Table I shows the number of experiments, reignition occurrences, and corresponding probabilities for current levels of 20 kA and above. As can be observed from the table, as the interrupted current increases, the probability of reignition also increases.

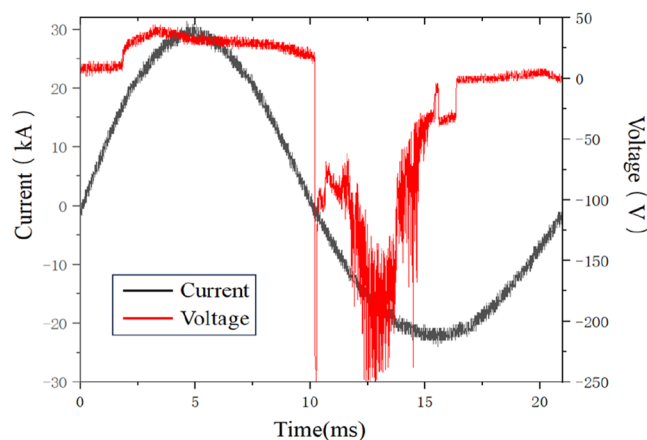
As shown in Fig. 4, for the six-slot cup-type contact at the beginning of the second half-wave in the experiments, the arc reignition position was almost always at the edge of the contact, and some cathode spots even spread to the side surface of the contact, where there is no corresponding magnetic field constraint. The arc jets generated from the side surface can elongate the arc channel and increase the

TABLE I. The number and probability of reignition.

Current (kA)	Number of experiments	Number of reignitions	Probability of reignition
20	4	1	25%
25	4	2	50%
30	4	4	100%

voltage noise. This increases the difficulty for the cathode spot to spread across the entire contact surface and also leads to an uneven magnetic field distribution, weakening the arc control capability.

Figure 5 shows the typical voltage and current waveform of cup-type contacts at the 30 kA current level. It can be observed that in the early period after the current zero, the voltage exhibits multiple sharp

**FIG. 4.** The arc reignition position of cup-type contacts at the beginning of the second half-wave.**FIG. 5.** Typical voltage and current waveforms of 30 kA interruption experiment.

fluctuations accompanied by severe noise, and then, the voltage gradually returns to stability after 6 ms.

By systematically examining all waveform data in the 25–30 kA range and analyzing the voltage waveforms in conjunction with the arc evolution recorded simultaneously by the two high-speed cameras, the reignition process in the second half-wave of the interruption tests can be divided into the stages shown in Fig. 6. On the other hand, because the number of tests at each current level is limited and the voltage exhibits strong fluctuations during the reignition period, it is difficult to make a precise comparison of absolute voltage values between different current levels. Therefore, in this part of the analysis, we mainly focus on the characteristics of each stage and the overall evolution trends. The descriptions of the five reignition stages are given as follows:

- (1) **Reignition stage:** At the beginning of the reignition half-wave, with the injection of reverse current, the voltage surges abruptly and the arc reignites. During the dielectric recovery phase, the remaining plasma between the contacts is weak and chaotic, and due to hysteresis, there is still a residual magnetic field at the current zero, which drives the continuous movement of the plasma. At this point, the disturbing effect of the transverse magnetic field on the plasma is greater than the stabilizing effect of the axial magnetic field on the plasma; the plasma randomly searches for a new ignition location, often in unconditioned areas such as the conductive rod or the side of the contact.
- (2) **Disordered movement stage:** During this stage, the cathode spots undergo short-term spreading motion along the edge and side surface of the contact, with occasional single jet appearing on the side. The arc voltage remains generally at a lower level throughout this period. Meanwhile, the cathode spots gradually migrate back to the surface of the contact plate, leading to a progressive shortening of the arc channel. Toward the later part of the S2 stage, the arc voltage decreases further.
- (3) **Dispersed arc stage:** In this stage, as the current gradually increases and the contact gap continues to widen, the arc channel lengthens correspondingly, resulting in a progressive rise in arc voltage. Meanwhile, the plasma density between the contacts increases steadily. There is no distinguishable plasma jet at this stage. The inter-electrode plasma remains in a dispersed state, rotating between the electrodes under the influence of the magnetic field, and the voltage noise is larger than that in the disordered movement stage.
- (4) **Transition stage:** As the AMF continuously strengthens, the confining effect of the AMF on the arc becomes greater than the disturbing effect of the transverse magnetic field component generated by the contacts, at which point the arc state starts to transition from the dispersed arc state to the diffused arc stage with clearer plasma jets, the plasma rotation between electrodes gradually slows down, and the arc transitions into a diffused mode. The arc voltage and noise also gradually decrease.
- (5) **Diffused arc stage:** After the current peak, the AMF can effectively control the arc, producing clear jets. The inclined angle of the jets becomes smaller, shortening the arc path between cathode and anode. The cathode spots continuously spread outward until they cover nearly the entire contact plate surface. The arc voltage returns to a stable and low level, with minimal noise, and the arc gradually extinguishes at the second current zero.

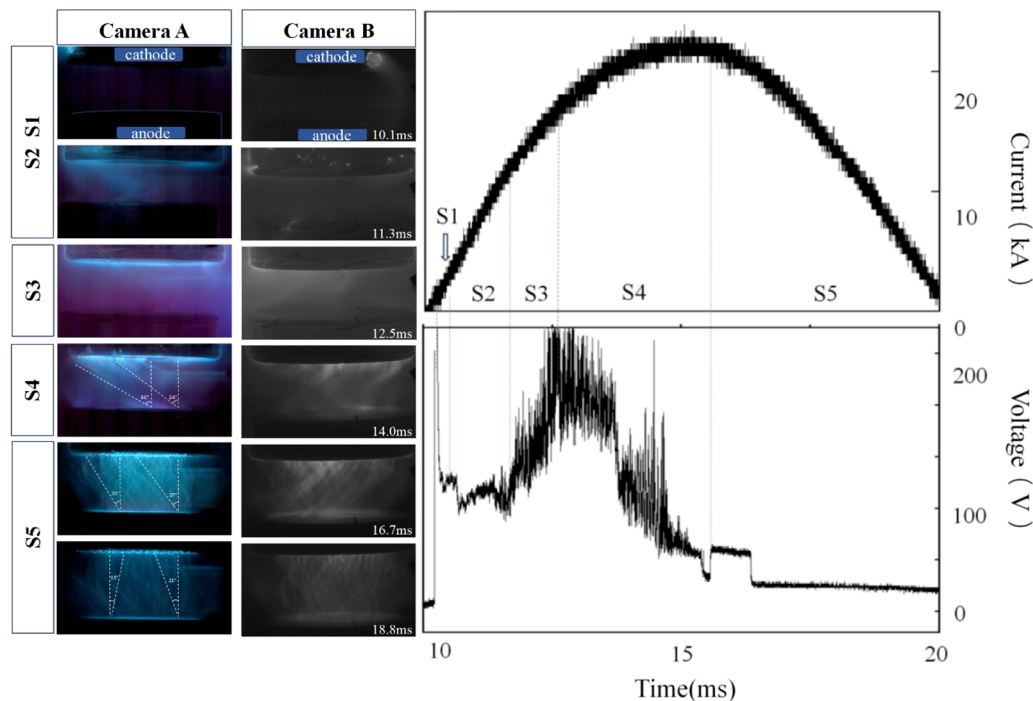


FIG. 6. The reignition stages in the second half-wave of the interruption experiment. The peak value of arc current in the first half-wave was 29.8 kA and inclination angles of the arc jets and the corresponding arcing times in stages S4 and S5.

Although the arc morphology and arc-voltage fluctuations vary to some extent among repeated reignition tests, the voltage waveforms of the cup-type contacts still exhibit similar evolutionary trends at each stage. As the peak current increases from 25 to 30 kA, the duration of the early unstable period (including the reignition and disordered-motion stages) tends to become shorter. This is because, at higher current levels, the axial magnetic field is established more rapidly and the cathode spots return to the contact surface earlier. At the same time, Fig. 6 also shows the inclination angles of the arc jets in stages S4 and S5, measured with respect to the normal of the contact surface. As the current rises toward the peak of the second half-wave (stage S4), B_z is enhanced and the plasma begins to form well-defined jet structures; the central axes of the jets exhibit inclination angles of approximately 54° – 60° , and the arc column is significantly elongated and tilted. After the current peak (stage S5), B_z reaches its maximum, and the jets are gradually “straightened,” with the inclination angle reduced to about 30° – 37° , and subsequently further decreased to 15° – 21° . In this regime, the arc path is markedly shortened, the jets become more concentrated, and the arc evolves into a stable multi-jet diffused mode. By using these approximate angle ranges, the evolution in stage S5 can be quantified: as the jet angle becomes smaller, the arc path is shortened, the effective arc resistance is reduced, and the arc voltage at a given current is lowered, which is consistent with the experimentally observed decrease in voltage level and noise.

These five stages encompass various phenomena of reignition during the second half-wave in interruption experiments. From these stages, the process of reignition and the main reasons for voltage instability and intense noise can be observed clearly. In order to further

investigate the driving mechanisms of plasma motion during the second half-wave, a three-dimensional magnetic field simulation model was established to analyze the magnetic field distribution on the contact surface under different operating conditions. The current, contact gap, and plasma column are shown in Fig. 7. The peak current is set to 30 kA.

In the simulation model, the time-varying eddy-current solver in ANSYS Maxwell is used. The input current is a 50 Hz LC oscillation waveform consistent with the experiments, and time-stepping starts from the first half-wave, so that the axial component (B_z) and the axial component (B_z) evolve continuously in time. It is first assumed that the plasma arc column uniformly occupies the contact gap, and B_z and B_t of the magnetic field on the moving contact surface are calculated, respectively. Figure 8 illustrates the evolution of both B_z and B_t distributions on the contact surface during the arcing process from 10 to 20 ms in the second half-wave. The case of B_z , which is defined as positive when pointing outward, perpendicular to the plane of the page, is straightforward to analyze. However, B_t is more complex due to the lack of a unified direction in the contact surface plane. To simplify and clearly visualize the B_t distribution, the projection of the synthetic magnetic vector onto the contact plane is used instead of analyzing the x and y components separately. This method allows for a more intuitive representation of the B_t behavior.

Figure 9(a) presents the variation of the maximum magnitude of B_z and B_t with gap distance, which reflects the impact of the magnetic field on arc behavior. It can be observed that the peak values of both B_z and B_t first increase and then decrease with the increasing gap, which correlates well with the variation trend of arc current during the

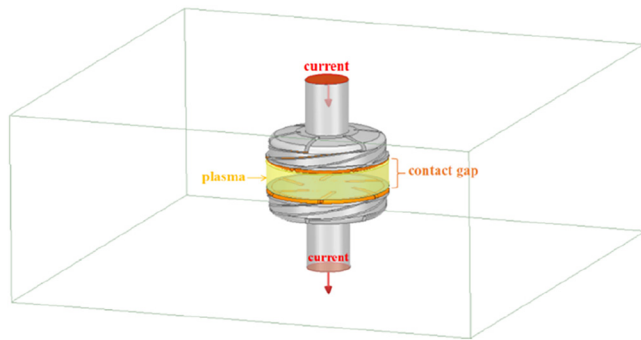


FIG. 7. Three-dimensional magnetic field simulation model (arc diffused across the entire contact surface).

second half-wave. Moreover, the B_z peak remains consistently higher than that of the B_t throughout the second half-wave, which is attributed to the cup-shaped structure of the AMF contact. At the current peak of the second half-wave, the difference between B_z and B_t reaches its maximum, indicating the strongest calibration effect of the B_z on the arc. This enhances the control over the arc transition to the diffused mode and reduces the plasma dispersion and inter-electrode rotation.

As shown in Fig. 8, assuming that the plasma is uniformly distributed across the entire contact surface, the magnetic field distribution—both axial and transverse—exhibits a high degree of symmetry, primarily influenced by the six slots of the contact plates. For the B_z , its peak value remains predominantly at the slot positions throughout the evolution process. This is because the slots can effectively suppress the eddy-current effects on the contact surface, thereby reducing the counteractive interference on the main magnetic field and resulting in localized field enhancement—termed as the “magnetic field peak zone.” As the current increases (from 13 to 16 ms, with the gap distance expanding from 26 to 32 mm), the B_z strength intensifies significantly, and the peak zone gradually diffuses outward, expanding the influence of the B_z and forming a more uniform axial field distribution. During this phase, the B_z exerts a distinct axial control and confining effect on the plasma, which helps suppress arc instability and facilitates the transition of the arc into a diffused mode.

As for the B_t , it can be observed from Fig. 8 that during the low-current stages (10–12 and 19–20 ms), the B_t vectors are mainly concentrated around the slotted regions, showing higher magnitudes near each slot. At the high-current stage (13–17 ms), the field directions progressively overcome the disturbances induced by the contacts and residual magnetic field, evolving into a counterclockwise swirling pattern. Furthermore, the higher-intensity B_t begins to concentrate toward the contact periphery at the high-current stage. However, with the concurrent strengthening of the B_z , the perturbative effect of the B_t gradually diminishes after 16 ms, and the B_z becomes the dominant magnetic field component.

At the beginning of the second half-wave, the peak value of B_t is nearly equal to that of B_z , and the B_z has not yet become dominant. To quantitatively characterize this competition between axial and transverse fields, we introduce a dimensionless ratio derived from the 3D magnetic field simulations

$$R_B = |B_t|/|B_z|,$$

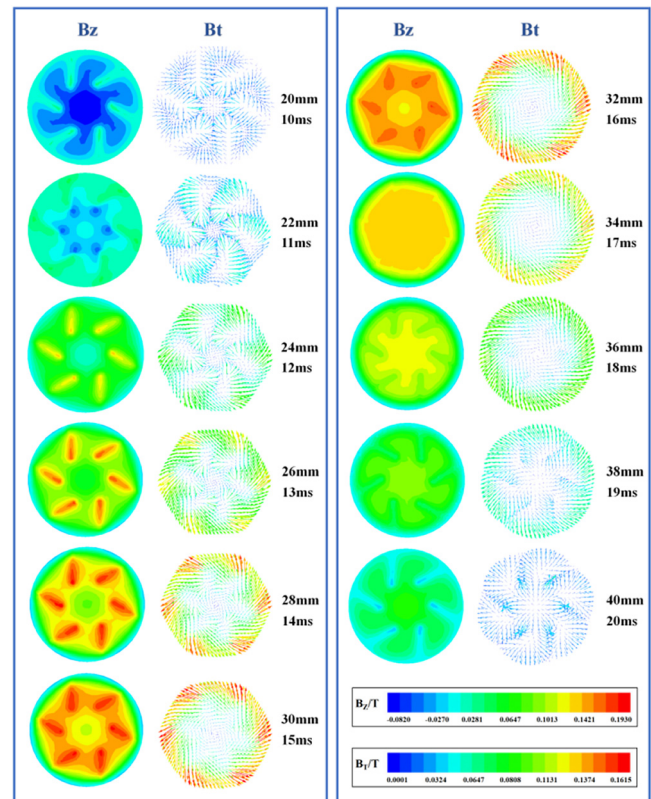


FIG. 8. Evolution of magnetic field distribution on the contact surface during the second half-wave.

where B_t and B_z denote the absolute values of the transverse and axial components on the contact surface during the second half-wave. The variation of R_B with the contact gap is plotted in Fig. 9(b) and is used to quantify the arc-controlling capability of the composite magnetic field. The simulation results show that in the early reignition stage (i.e., $g \approx 20$ –22 mm, $t \approx 10$ –11 ms), the transverse field is comparable to or even as strong as the axial field in the resultant field. In this regime, the residual axial field at current zero is not strong enough to effectively confine the plasma, while B_t has a more pronounced impact on the local $J \times B$ force. As a result, the plasma is prone to rotation and lateral displacement, and the arc is highly unstable during the initial reignition stage, which manifests as strong voltage noise. As the current increases and the contact gap widens, the amplitude of B_z grows faster than that of B_t , and R_B gradually drops below unity, reaching about $R_B \approx 0.8$ near the peak of the second half-wave (i.e., $g \approx 30$ mm). At this point, the axial field becomes dominant in the resultant field, the axial confinement of the plasma is significantly enhanced, and the arc progressively transitions from a B_t -perturbed dispersed state to a B_z -dominated diffused arc mode, accompanied by a marked reduction in voltage noise.

The earlier qualitative description is further supported and refined by these quantitative results: during the initial reignition stage, when $R_B \approx 1$, the axial component B_z is not yet sufficient to provide effective confinement and the transverse component B_t dominates the

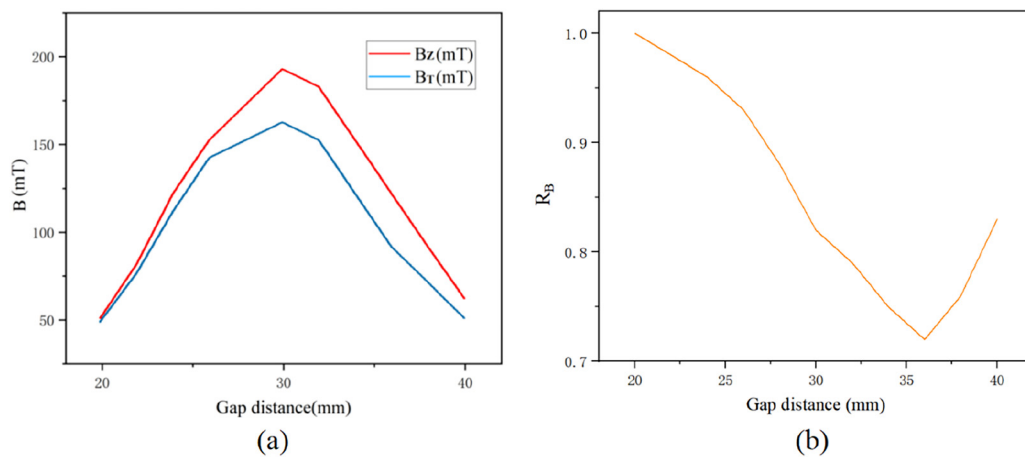


FIG. 9. The variation of the absolute peak values of both the B_z , B_r (a), and R_B (b) with increasing contact gap.

arc dynamics; as $R_B < 1$ and B_z continues to increase, the arc gradually evolves into a stable diffused state dominated by B_z . At the same time, the introduction of R_B provides a useful quantitative benchmark for comparing the present AMF contact design with other AMF configurations.

In the above analysis, a uniformly distributed plasma in the entire inter-contact region is assumed as an ideal condition. However, in practical interruption experiments, particularly during the initial stage of reignition under a high-current arcing test with a peak current of 30 kA, the plasma is unlikely to occupy the entire inter-contact region. The arc typically initiates at the edge of the contact plate or on the lateral surface of the contact, producing a jet-like arc column as shown in Fig. 6(S1). Therefore, corresponding magnetic field simulations are conducted for these two commonly observed reignition scenarios.

Figure 10 illustrates the simulation model of the edge-triggered reignition scenario, in which a plasma column with a diameter of 20 mm is assumed to be tangent to the edge of the contact plate, while other boundary conditions remain unchanged. Figure 11 shows the distributions of B_z (top) and B_r (bottom), on the cup-type AMF contact surface at the early stage of the second half-wave (10 and 11 ms), in which the initial arc presents remarkable instability closely related to the magnetic field distribution. At 10 ms, the B_z is generally weak, relying primarily on residual magnetism concentrated around the center and slot positions. Although the slots exhibit slight local enhancement, no distinct magnetic field peak zone has yet formed, indicating that the confining and control effect of B_z on the plasma has not been established at this moment, making it difficult to effectively restrain the arc. By 11 ms, with the increase in current, the B_z strength significantly intensifies but becomes more nonuniform. The magnetic field peak becomes concentrated near a single slot, forming a localized “magnetic peak zone,” which, however, lies outside the plasma column area, making it ineffective in governing the plasma expansion across the entire contact surface.

In contrast, the B_r distribution at 10 ms appears relatively uniform, with the residual magnetic field vectors radiating outward from the slots in a well-organized pattern and exhibiting limited perturbation. At 11 ms, the B_r vectors increase significantly in magnitude along

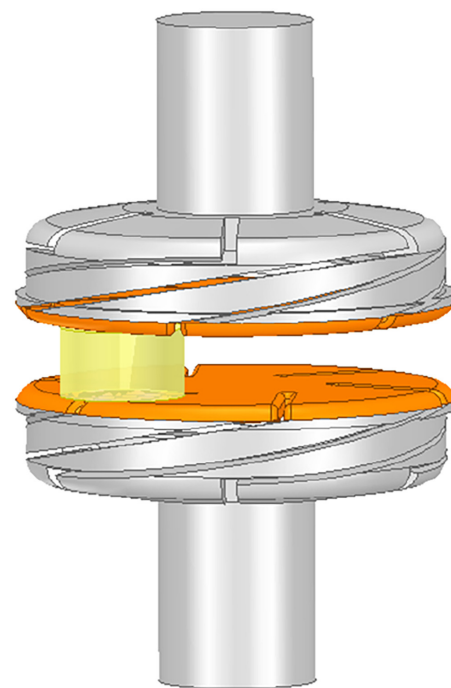


FIG. 10. The simulation model for edge-initiated reignition.

with the current, and the B_r peak even surpasses the B_z peak. A pronounced swirling structure and intense magnetic aggregation emerge around the arc column, forming a distinct vortex-type distribution. Notably, the B_r amplitude reaches its maximum near the contact center.

Overall, the edge-triggered reignition pattern during the reignition phase leads to uneven B_z and B_r distributions. The B_z is weakened in magnitude, and the “magnetic peak zone” becomes reduced in area. Meanwhile, the amplitude of the B_r is locally enhanced near the arc

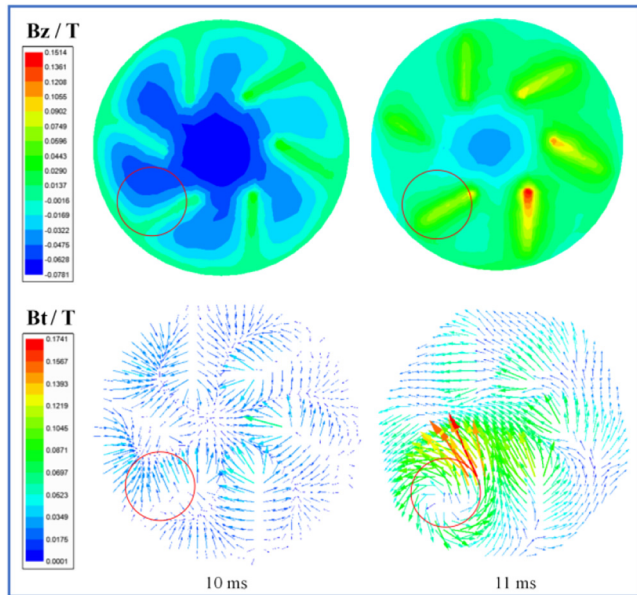


FIG. 11. The distribution characteristics of the B_z and B_t on the surface of the contact at the early stage of the second half-wave under edge-initiated reignition conditions.

column. The asymmetry and localized concentration of the magnetic field introduce significant disturbances to the plasma, which may result in arc displacement, unstable elongation, and even rapid rotational motion between the contacts. These phenomena indicate that the arc is in a highly unstable state during the initial reignition phase. Whether the arc can ultimately stabilize into a diffused mode depends on whether the B_z can reassert dominance in subsequent phases and provide continuous controlling influence.

Figure 12 shows the simulation model of the arc jet column ignited at the contact side surface. Based on actual observations from experiments, a plasma arc column with a diameter of 14 mm is established to make contact with the side surfaces of both the moving and fixed cup-type contacts. A circular computational region with a diameter of 280 mm is used to calculate the magnetic field distribution on the surface of the moving contact and its surrounding plane, while all other simulation conditions remain unchanged.

Figure 13 presents the magnetic field distribution on the moving contact surface and its surrounding plane at the initial reignition moment of 11 ms. The left panel shows the B_z distribution, while the right panel displays the synthesized B_t vector distribution. In the axial field map, due to the side-ignition of the arc, the B_z exhibits a pronounced asymmetry. The original centrosymmetric pattern formed by the six-slot structure is significantly disturbed. As shown in Fig. 13, during the second half-wave, the arc column generates current and circumferential magnetic fields at the intersection with the plane of the moving contact surface, as indicated by the arrows. When the axial component of the magnetic field B_z is projected onto the contact plane, a “peak/anti-peak” high-gradient zone is formed within the arc column region as shown in Fig. 13. At the same time, strong localized peaks of B_z also appear on the contact surface as in the edge-initiated reignition conditions as shown in Fig. 11. Such an asymmetric B_z distribution

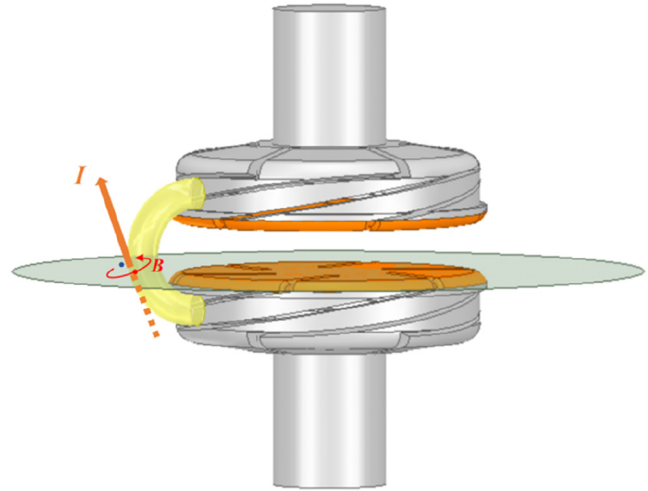


FIG. 12. The simulation model of a plasma jet column initiated from the side surface of the contact and the schematic diagram of current and magnetic field directions.

weakens its inherent confining capability on the arc. Instead of promoting arc stabilization, it constrains the arc’s radial diffusion toward the contact gap and may lead to localized arc contraction, thereby impairing the effectiveness of B_z in stabilizing the plasma during the initial arcing stage.

The B_t vectors exhibit a clear vortex-like rotational structure surrounding the arc jet. Particularly near the outer edge of the contact cup, vector arrows increase in magnitude and undergo directional twisting, indicating significant local amplification of the B_t . The region where the arc jet is located generates an asymmetric composite magnetic field that strongly perturbs the plasma. Under this side-jet condition, combining the current waveform with the arc cross section, the current density and $J \times B$ force in the simulation model are evaluated. For a plasma arc column with a diameter of 14 mm carrying a second half-wave peak current of 20–25 kA, the resulting electromagnetic force density is on the order of 10^7 – 10^8 N/m³. Such a force level can noticeably drive radial contraction of the plasma, while also causing rapid plasma acceleration, leading to strong unstable perturbations, lateral motion, or rotational behavior.¹⁹ Consequently, under this operating condition, the resulting intensified magnetic force exerts substantial traction on the arc plasma, leading to arc channel oscillations, unstable elongation, and even sideward arc displacement, thereby exacerbating the instability of the arc during the initial reignition phase.

Taken together, these two figures demonstrate that under side-triggered arc jet conditions, the original symmetric structures of both B_z and B_t are severely disrupted, exhibiting strong asymmetry and local enhancement. Under such ignition scenarios, the arc behavior becomes highly dependent on whether the B_z can subsequently reestablish a uniform distribution over a large area and overcome localized B_t disturbances, enabling the transition from a highly asymmetric, perturbed state to a stable, diffused arc mode.

IV. CONCLUSION

The paper conducts drawn vacuum arc experiments with 126 kV AMF contact, especially focusing on the reignition phenomenon and

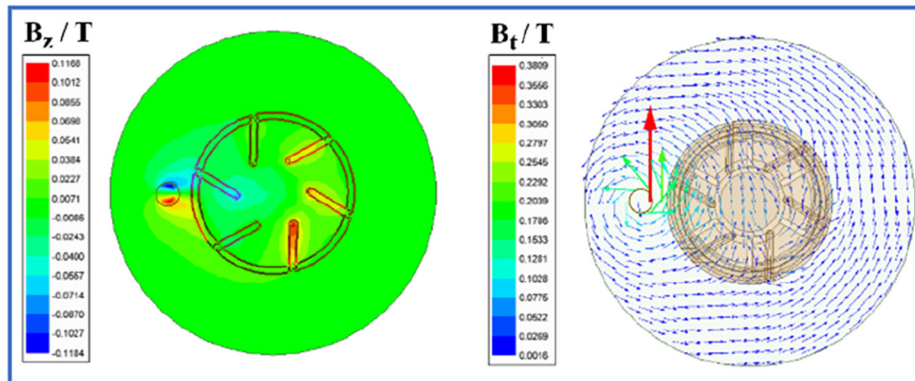


FIG. 13. The distribution characteristics of the B_z and B_t on the surface of the contact at the early stage of the second half-wave under the plasma jet column condition and “peak/anti-peak” high-gradient zone.

evolution of the arc in the second half-wave during high-current interruption. The main conclusions of the paper are as follows:

- (1) Through experimental research, it has been found that the evolution characteristics of the arc after reignition in the current interruption process of vacuum circuit breakers can be divided into five stages: in the reignition stage, magnetic field control has not yet been established, and the arc is extremely unstable; during the disordered movement stage, cathode spots wander locally and the arc path changes frequently; in the dispersed arc stage, the plasma density gradually increases but no stable arc jet is formed; in the transition stage, the current increases and the axial magnetic field strengthens, gradually gaining dominance and stabilizing the arc; finally, in the diffused arc stage, the arc expands across the entire contact surface under strong B_z control, forming a stable multi-jet diffusion pattern and creating favorable conditions for final current interruption.
- (2) Under the ideal condition that the arc fully diffuses across the contact surface, the simulated magnetic field exhibits high symmetry. At the beginning of the reignition process, the B_z shows relatively low magnitude and does not dominate the overall magnetic field, making it difficult to effectively control the arc. As the current increases, the calibration effect of the B_z is continuously enhanced. At the same time, the B_t remains weak and evenly distributed between slots, resulting in mild disturbance. This creates a magnetic field environment that is more conducive to stable arc control and smooth transition.

The reignition position of the cup-type contact is more likely to occur at the lateral side and the edge of the contact plate, which significantly disrupts the originally symmetrical magnetic field distribution. The B_z exhibits strongly asymmetric peaks, failing to provide effective axial confinement. Meanwhile, the B_t is markedly intensified in local regions. Such asymmetry tends to induce arc displacement, elongation, or unstable jumping, thereby hindering the arc's transition into a stable diffused mode and resulting in significant voltage fluctuations and pronounced noise during the initial stage of reignition.

In summary, the evolution of the arc during reignition is thus highly sensitive to the arc initiation location, magnetic field configuration, and current magnitude. Basically, the arc initiation position during reignition is influenced by the arc evolution in the preceding half-wave and is closely related to factors such as the surface condition of the contacts and the degree of conditioning of the vacuum interrupter.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yiming Ding: Investigation (lead); Visualization (equal); Writing – original draft (lead). **Zongqian Shi:** Conceptualization (lead); Supervision (lead); Writing – review & editing (lead). **Yongpeng Mo:** Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Ruiqi Liu:** Investigation (equal). **Junhao Niu:** Investigation (equal). **Benfu Zhu:** Investigation (equal). **Qinyuan Xian:** Investigation (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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